EXPERIMENT 9

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-- Name: VANSHIKA DIXIT
-- Roll No: 2401010010
-- Topic: GRANT, REVOKE, SAVEPOINT
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```

Consider the following tables namely "DEPARTMENTS" and "EMPLOYEES" Their schemas

are as follows,

Departments (dept no , dept name , dept_location); Employees (emp_id , emp_name , emp_salary,dept_no);

a) Develop a query to grant all privileges of employees table into departments table

b) Develop a query to grant some privileges of employees table into departments table

c) Develop a query to revoke all privileges of employees table from departments table

d) Develop a query to revoke some privileges of employees table from departments table

e) Write a query to implement the save point.

```
CREATE DATABASE IF NOT EXISTS Assignment9_DB;
USE Assignment9_DB;

CREATE TABLE Departments (
   dept_no INT PRIMARY KEY,
   dept_name VARCHAR(50),
   dept_location VARCHAR(50)
);

CREATE TABLE Employees (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   emp_salary DECIMAL(10,2),
   dept_no INT
```

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);

INSERT INTO Departments VALUES
(1, 'HR', 'Delhi'),
(2, 'Finance', 'Mumbai');

INSERT INTO Employees VALUES
(101, 'Vanshika', 40000, 1),
(102, 'Roshani', 50000, 2);

SELECT * FROM Departments;
SELECT * FROM Employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no
101	Vanshika	40000.00	1
102	Roshani	50000.00	2
NULL	NULL	NULL	NULL

```
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123';

GRANT ALL PRIVILEGES ON Assignment9_DB.Employees TO 'user1'@'localhost';

GRANT SELECT, INSERT ON Assignment9_DB.Employees TO 'user1'@'localhost';

REVOKE ALL PRIVILEGES ON Assignment9_DB.Employees FROM 'user1'@'localhost';

REVOKE INSERT ON Assignment9_DB.Employees FROM 'user1'@'localhost';

START TRANSACTION;

UPDATE Employees SET emp_salary = emp_salary + 5000 WHERE emp_id = 101;
```

SAVEPOINT sp1;

UPDATE Employees SET emp_salary = emp_salary + 3000 WHERE emp_id = 102;

ROLLBACK TO sp1;

COMMIT;

SELECT * FROM Employees;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no
101	Vanshika	45000.00	1
102	Roshani	50000.00	2
NULL	NULL	NULL	NULL

FULL HISTORY

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:28:13	CREATE DATABASE IF NOT EXISTS Assignment9_DB	1 row(s) affected	0.015 sec
2	10:28:13	USE Assignment9_DB	0 row(s) affected	0.000 sec
3	10:28:24	CREATE TABLE Departments (dept_no INT PRIMARY KEY, dept_name VARCHAR(50), dept_location VAR...	0 row(s) affected	0.046 sec
4	10:28:38	CREATE TABLE Employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50), emp_salary DECIMAL(...	0 row(s) affected	0.031 sec
5	10:28:52	INSERT INTO Departments VALUES (1, 'HR', 'Delhi'), (2, 'Finance', 'Mumbai')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.032 sec
6	10:29:33	INSERT INTO Employees VALUES (101, 'Vanshika', 40000, 1), (102, 'Roshani', 50000, 2)	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.016 sec
7	10:29:49	SELECT * FROM Employees LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
8	10:30:37	CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123'	Error Code: 1396. Operation CREATE USER failed for 'user1'@'localhost'	0.032 sec
9	10:30:50	GRANT ALL PRIVILEGES ON Assignment9_DB.Employees TO 'user1'@'localhost'	0 row(s) affected	0.016 sec
10	10:31:10	GRANT SELECT, INSERT ON Assignment9_DB.Employees TO 'user1'@'localhost'	0 row(s) affected	0.015 sec
11	10:31:48	REVOKE ALL PRIVILEGES ON Assignment9_DB.Employees FROM 'user1'@'localhost'	0 row(s) affected	0.031 sec
12	10:32:02	REVOKE INSERT ON Assignment9_DB.Employees FROM 'user1'@'localhost'	Error Code: 1147. There is no such grant defined for user 'user1' on host 'localhost' on table 'employees'	0.000 sec
13	10:32:21	START TRANSACTION	0 row(s) affected	0.000 sec
14	10:32:35	UPDATE Employees SET emp_salary = emp_salary + 5000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
15	10:32:56	SAVEPOINT sp1	0 row(s) affected	0.000 sec
16	10:33:07	UPDATE Employees SET emp_salary = emp_salary + 3000 WHERE emp_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
17	10:33:19	ROLLBACK TO sp1	0 row(s) affected	0.015 sec
18	10:33:31	COMMIT	0 row(s) affected	0.016 sec
19	10:33:53	SELECT * FROM Employees LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec